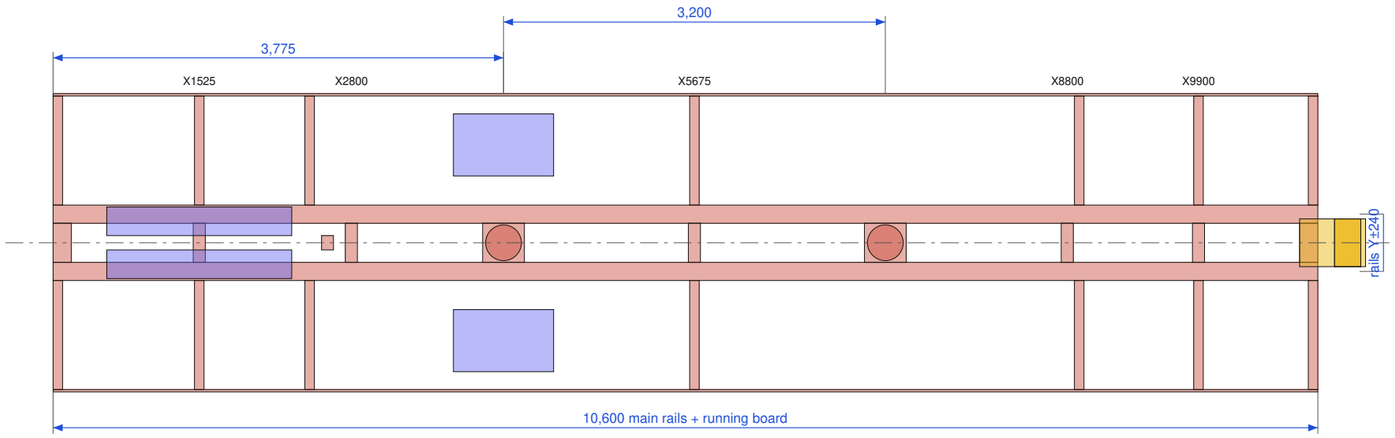
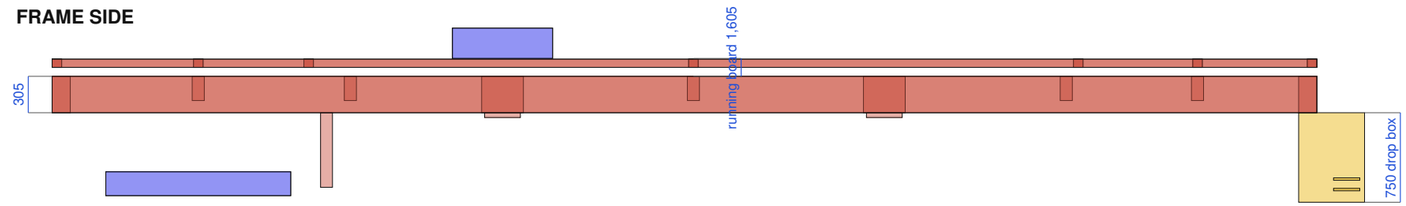
FRAME PLAN (deck plate removed)



FRAME SIDE



MEMBER SCHEDULE

#	Part	Section / material	Qty	Length	Location
1	Main rail	HSS12x6x3/8 A500C	2	10,600	Y±240, Z1150-1455 (LT18)
2	End sill	HSS12x6x3/8 A500C	2	328	X300 / X10,748
3	Crossmember	HSS8x4x1/4 A500C	5	328	X1525/2800/5675/8800/9900
4	Body bolster box	PL12 flg 350 + PL10 web	2	328	X4075 / X7275
4a	Running-board outrigger	C3x4.1 A36	14	914	X340/1525/2450/5675/8900/9900/10860
4b	Board edge angle	L3x3x1/4 A36	2	10,600	Y±1,240
4c	Swing-arm anchor	PL20 box	1	625	X2600, leading truck
5	Centre plate	PL40 Ø300 + Ø60 pin bore	2	-	under bolster
6	Drop drawbar box	PL16 A572-50 box 552x400	1	750 tall	pin X11,150 Z550
7	Knee gussets	PL16 A572-50	4	600	box to each rail
8	Running board / deck	PL6 tread plate	1	10,600x2,500	top Z1,605
9	Ballast pocket	PL20 ends / PL12 sides	2	840x520x255	over front bolster

NOTES

Hand checks (calc notes C4, C4b, C5, C8):
rails 90 MPa at 2 g; 225 MPa at 200 kN drawbar (< Fy 345).
bolster 34 MPa at 2 g; drop box 34 MPa / weld 85 MPa at 200 kN.
All crossmembers cope-fit, 6 mm fillet all round.
Rails: no splices. Camber none. Check diagonals within 3 mm.
Lifting: 4 jacking pads under bolster ends (load 2 g).
Equipment hard-points to be added from AIE/Poclain data (AWAIT).

PE STAMP / SIGNATURE

(blank - for reviewing NV PE)

UNLESS NOTED: ISO 2768-mK general tolerances
Linear ±0.5 (≤120) / ±0.8 (≤400) / ±1.5 (>400)
Fab. cut ±1.5 · holes pos. ±0.8 · angles ±0.5°
Machined: Ra 3.2 · bores H7/H8 as noted · break edges
GD&T ASME Y14.5-2018 · welds AWS A2.4 / D1.1

PULSIPHER FARMS CHRISTMAS TRAIN — PRELIMINARY, NOT FOR CONSTRUCTION UNTIL PE STAMPED

PULSIPHER FARMS · LOGANDALE, NEVADA · CHRISTMAS TRAIN (PF-CT)

Locomotive frame and member schedule

Sheet L-102 · REV L2 · 2026-10-27 · 1:50 at 17x11

Units mm unless noted · Z above top of rail · gauge 914.4 · 3rd-angle

Drawn: owner team 2026-10-27
Checked: _____ Date: ____
Approved (PE): _____